
Daimler AG

Common Powertrain Controller CPC4-ECAN



Operating Manual

Revision 1.4
2018-12-12

Software Release R53
Diagnostic Version 0708

Version History

Date	Revision	Changes
2018-12-12	1.4	- Chapter 3.6.4. (Connector ST4): Input DI_FLEX_19 corrected to switch to battery voltage - Chapter 3.8.5.3. (Fault detection): In example table and following text for exemplary fault detection parameter 35/25 (3 07 DO Fault Detection) the values 1 and 9 are marked bold - Chapter 4.1. (SAE J1939 Transport Protocol) shifted from 4.2. - Chapter 4.2. (DI_FLEX Inputs) renamed to "Digital Inputs" - Chapter 4.2.1. renamed to "Specification for DI_FLEX Inputs" - Chapter 4.3. (Vehicle Speed Sensor) added - Chapter 4.4. (Driver Outputs) added - Chapter 4.5. (Comparison of corresponding pins) added - Chapter 5.1.1. (PGR001 – Communication) and 6.1. (List of Parameters): Parameter 001/86 (Prop message for KWP Gateway) added - Chapter 5.1.3. (PGR003 – Common Limiters) and 6.1. (List of Parameters) and 7.1.1.5. (Speed limitation in Torque Control Mode): Parameter values 1, 2, 3 added for parameters 003/08 and 003/21 - Chapter 5.1.5. (PGR005 – Limiters LIM0 and LIM1) and 6.1. (List of Parameters) and 7.1.1.5. (Speed limitation in Torque Control Mode): Parameter values 1, 2, 3 added for parameters 005/01 and 005/02 - Chapter 5.1.6. (PGR006 – Limiters AC/LIM2) and 6.1. (List of Parameters) and 7.1.1.5. (Speed limitation in Torque Control Mode): Parameter values 1, 2, 3 added for parameter 006/01 - Chapter 5.1.11. (PGR011 – Accelerator Pedal) and 6.1. (List of Parameters): Default value of parameter 11/46 changed from 100 % to 80 %Chapter 5.1.19. (PGR019 – Automatic Fan Activation) and 6.1. (List of Parameters): Value 251 (Fan defined in MR2) added for parameter 19/28 and set as default, default value of parameter 35/35 changed from 0 to 9, 35/33 from 0 to 1 and parameter equivalents added - Chapter 5.1.22 (PGR022 – Limiter Governor) and 6.1. (List of Parameters) and 7.1.1.5. (Speed limitation in Torque Control Mode): Parameter values 0, 1, 2, 3, 4 added for parameter 22/19 and default value changed from 6 to 5, parameter values 1, 2, 3 added for parameters 22/18 and 22/20, parameter values 1, 2, 3 added for parameter 22/21 and default value changed from 4 to 0 - Chapter 5.1.35. (PGR035 – Digital Outputs) and 6.1. (List of Parameters): Default value of parameters 35/20, 35/21, 35/22, 35/23, 35/27, 35/28, 35/29, 35/30, 35/31, 35/34, 35/35, 35/36, 35/37 changed from 0 to 9, default value of parameters 35/32 and 35/33 changed from 0 to 1 for activation of output pin fault detection - Chapter 6.1. (List of Parameters): Parameter equivalent values corrected for parameter pairs ADM3 02/03 – CPC4-ECAN 35/08, ADM3 02/03 CPC4-ECAN 35/15, ADM3 02/05 – CPC4-ECAN 35/07, ADM3 02/06 – CPC4-ECAN 35/06, ADM3 13/14 – CPC4-ECAN 13/01, ADM3 13/15 – CPC4-ECAN 13/13, ADM3 13/16 – CPC4-ECAN 13/09, ADM3 13/16 – CPC4-ECAN 13/10, ADM3 13/18 – CPC4-ECAN 13/12, ADM3 14/10 – CPC4-ECAN 35/16, Equivalent parameters for ADM3 17/10 and 17/11 corrected to 017/29 and 017/30 respectively - Chapter 7.1.1.6. (Switchable Torque Demand) added - Chapter 7.2.2. (Transmission Neutral / Driveline status): J1939 input signal "Transmission Driveline Engaged " from ERC1 corrected to ETC1 - Chapter 7.8.2.3. (Engine Brake Operation via J1939 CAN Message EBC1): Parameter number corrected for 013/23 (Engine Brake Switch Config) - Chapter 7.8.4.1. (Anti-Lock Brake (ABS) Intervention and the Engine Brake): Parameter value for input pins DI_FLEX_22 und DI_FLEX_19 corrected to "enable ABS input" - Chapter 8.2.2. (Fault Codes) renamed to "Fault Codes for CPC4-ECAN and MR2": MR2 fault codes added to list, Fault code SPN 629 FMI 14 (MR2 Diagnosis Version Mismatch) added, replaces 625/8 and 625/16
2018-07-06	1.3	- Chapter 3.3. (Wiring diagram of CPC4-ECAN): Function of pin 2/15 changed to "EngineBrakeHigh"; Function of pin 4/06 changed to "Misc (BatteryChargeIndicatorLamp)" - Chapter 3.6.2. (Connector ST2): Function of pin 2/15 corrected to "Misc. (e. g. Engine Brake High)"; Description of parameter 13/10 value 1 corrected to "engine brake high"; Calculation of engine brake stage out of DIFLEX13 and DIFLEX14 input values corrected - Chapter 3.6.4. (Connector ST4): Function of pin 4/13 corrected to "Misc. (e. g. Enable Temposet)" - Chapter 6. (Parameter Mapping ADM3 – CPC4-ECAN): List of parameters now explained more clearly - Chapter 7.1.1.4. (Overspeed Governor / "Droop"): Table for engine speed range and droop depending on engine variant corrected - Chapter 7.1.2. (Speed Control Mode (PTO Control Mode)): References to other chapters corrected - Chapter 7.2.8. (Split Gear) added - Chapter 7.3.3. (Cold Start with Grid Heater) added - Chapter 7.4.7. (Variable vehicle speed limitation (VSL) CC-Limiter Mode (ECE-R89)):

		Refined hint for ECE-R89 regulation - Chapter 7.6.6.3. (Accelerator Pedal with Driving from Superstructure (DSS)): Function of input pin 3/18 corrected to "Misc (EnableAbsInput); Function of input pin 4/13 corrected to "Misc (EnableTempoSet) - Chapter 8.3.1. (Analog Values): Analog value 72 deleted
2018-06-06	1.2	- Chapter 2.1. (Introduction): Correction of engine series - Chapter 2.2. (Abbreviations): Misleading abbreviations from PTCAN variant eliminated - Chapter 3.3. (Wiring diagram of CPC4-ECAN): Pins 3/19, 3/20, 3/21 deleted; Function of pin 3/18 changed to "Misc (EnableAbsInput)"; Function of pin 4/13 changed to "Misc (EnableTempoSet)"; Function of pin 4/17 changed to "Misc (Clutch2)"; Function of pin 1/04 changed to "Misc (GroundIdleValidationSwitch)" - Chapter 3.6.1. (Connector ST1): Function of pin 1/05 changed to Misc (e. g. Ground Idle Validation Switch); Value 2 of parameter 35/02 for pin 1/05 changed to "not used" Function of pin 1/15 changed to "Shutdown Override" - Chapter 3.6.3. (Connector ST3): Value 10 of parameter 08/13 changed to "Output shaft speed from transmission (E-Can)"; Value 10 of parameter 13/12 for pin 3/18 changed to "not used"; Value 2 of parameter 13/15 changed to "not used"; Function of pin 3/07 changed to "Misc."; Function of pin 3/10 changed to "Misc. (e.g. TIER4i DEF Lamp)"; Pins 3/19, 3/20, 3/21 deleted - Chapter 3.6.4. (Connector ST4): Value 10 of parameter 13/14 for pin 4/13 changed to "not used"; Value 2 of parameter 13/15 for pin 4/17 changed to "not used" - Chapter 3.7. (Power Supply CPC4-ECAN): Chapter renamed (CPC4-ECAN instead of CPC); Figure 3.11. corrected for CPC4-ECAN/MR2 environment - Chapter 3.7.1. (Start-up and Shutdown behavior of CPC4-ECAN): Chapter renamed; Abstract for shutdown times corrected - Chapter 3.8.2.1. (Technical Data): Functions for pins 1/01 (DI_FLEX_01), 1/11 (DI_FLEX_08), 1/15 (DI_FLEX_05), 2/11 (DI_FLEX_10), 2/13 (DI_FLEX_12), 2/14 (DI_FLEX_13), 2/15 (DI_FLEX_14), 3/18 (DI_FLEX_22), 4/13 (DI_FLEX_19), 4/18 (DI_FLEX_18) corrected, pin 4/03 deleted - Chapter 3.8.2.3. (Fault detection): Links to figures 3.13 and 3.14 corrected - Chapter 3.8.5.1. (Technical Data): Functions for pins 1/05 (DO_LP_LS_01), 1/13 (DO_LP_FLEX_01), 3/07 (DO_HP_FLEX_01), 3/08 (DO_HP_FLEX_02), 3/09 (DO_HP_LS_01), 3/12 (DO_LP_FLEX_04), 3/16 (DO_LP_FLEX_05), 3/17 (DO_HP_LS_04), 4/06 (DO_LP_FLEX_06), 4/07 (DO_HP_LS_02), 4/09 (DO_HP_LS_03), 4/10 (DO_HP_FLEX_03) corrected; Pins 4/01, 4/02 deleted - Chapter 3.8.5.3. (Fault detection): Table with values for parameter 35/25 "3 07 DO Fault Detection" corrected - Chapter 3.8.6.1. (Technical Data): Functions for pins 3/05 (AO_01) corrected - Chapter 3.8.7. renamed to "Communication Interface (CAN) - Chapter 3.8.7.1. (Technical Data): Function for pin 2/04 changed to K-Line; Pins 3/19, 3/20, 3/21 deleted - Chapter 5.1.1. (PGR001 - Communication): Parameter 01/81 (Starter Lockout Evaluation) added - Chapter 5.1.2. (PGR002 - Vehicle Parameters I): Value 6 of parameter 02/09 changed to "Allison, Eaton Ultrashift ASW: MR2 start" - Chapter 5.1.8. (PGR008 - Vehicle Speed Sensor): Value 10 of parameter 08/13 changed to "Output shaft speed from transmission (E-Can)" - Chapter 5.1.13. (PGR013 - Inputs): Value 10 of parameter 13/12 changed to "not used"; Value 10 of parameter 13/14 changed to "not used"; Value 2 of parameter 13/15 changed to "not used" - Chapter 5.1.31. (PGR035 - Digital Outputs): Value 2 of parameter 35/02 changed to "not used" - Chapter 6.1. (List of parameters): Parameter mapping ADM3 02/06 value 1 to CPC4-ECAN 35/06 corrected to 1 ↔ 2; Parameter mapping ADM3 13/18 value 11 to CPC4-ECAN 13/12 corrected to 9 ↔ 11; Value 6 of parameter 02/09 changed to "Allison, Eaton Ultrashift ASW: MR2 start"; Value 10 of parameter 08/13 changed to "Output shaft speed from transmission (E-Can)"; Value 10 of parameter 13/12 changed to "not used"; Value 10 of parameter 13/14 changed to "not used"; Value 2 of parameter 13/15 changed to "not used"; Value 2 of parameter 35/02 changed to "not used"

		- Chapter 7.1.1.4. (Overspeed Governor / "Droop"): Engine variants eliminated - Chapter 7.1.3. (Characteristics of Governor Types) deleted - Chapter 7.2. (Vehicle Identification Number (VIN)) deleted - Chapter 7.2.1. (Transmission Type): Value 6 of parameter 02/09 changed to "Allison, Eaton Ultrashift ASW: MR2 start" - Chapter 7.2.2.2. (Clutch Switch): Clutch switch information from Daimler transmission corrected to E-CAN Considered feature DPF for clutch condition deleted; Value 2 of parameter 13/15 changed to "not used" - Chapter 7.2.6.1. (Transmission Learn in routine) deleted - Chapter 7.3.1.2. (Engine Start via CPC4-ECAN and MR2 in Parallel (hardwired) – recommended: Value 6 of parameter 02/09 changed to "Allison, Eaton Ultrashift ASW: MR2 start" - Chapter 7.3.1.3. (Engine Start via CPC4-ECAN and MR2 in Series (hardwired or CAN): Abbreviation of transmission control unit changed to TCU; Value 6 of parameter 02/09 changed to "Allison, Eaton Ultrashift ASW: MR2 start" - Chapter 7.3.1.5. (Engine Start via CPC4-ECAN without MR2 (hardwired an CAN)): Value 6 of parameter 02/09 changed to "Allison, Eaton Ultrashift ASW: MR2 start" - Chapter 7.4.4. (Cruise Control (general)): Function of pin 3/18 changed to "Misc (EnableAbsInput)"; Function of pin 4/13 changed to "Misc (EnableTempoSet)"; Value 10 of parameter 13/12 changed to "not used"; Value 10 of parameter 13/14 changed to "not used"; Pin 4/02 deleted from outputs - Chapter 7.4.5. (Cruise Control with Engine Brakes): Reference to parameter 15/19 deleted from Fig. 7.13 - Chapter 7.4.6. (Cruise Control when using a Stalk Switch): Function of pin 3/18 changed to "Misc (EnableAbsInput)"; Function of pin 4/13 changed to "Misc (EnableTempoSet)"; Value 10 of parameter 13/12 changed to "not used"; Value 10 of parameter 13/14 changed to "not used"; Description of parameter 13/14 corrected - Chapter 7.4.7. (Variable vehicle speed limitation (VSL) CC-Limiter Mode (ECE-R89)): Function of pin 3/18 changed to "Misc (EnableAbsInput)"; Function of pin 4/13 changed to "Misc (EnableTempoSet)"; Value 10 of parameter 13/12 changed to "not used"; Value 10 of parameter 13/14 changed to "not used" - Chapter 7.6.3. (Dual Channel Analog Accelerator Pedal): Parameter 13/64 corrected to 13/61 (4 14 AI Selection) and made visible in parameter list - Chapter 7.6.4. (Analog Accelerator Pedal with IVS): Function of pin 1/04 changed to "Misc (GroundIdleValidationSwitch)" - Chapter 7.6.5. (Analog Accelerator Pedal via J1939 CAN message EEC2): Function of pin 3/18 changed to "Misc (EnableAbsInput)"; Function of pin 4/13 changed to "Misc (EnableTempoSet)"; Value 10 of parameter 13/12 changed to "not used"; Value 10 of parameter 13/14 changed to "not used" - Chapter 7.6.6.2. (Analog Remote Accelerator Pedal (hardwired or CAN): Reference to IVS1 and IVS2 deleted; Parameter list corrected - Chapter 7.6.6.3. (Accelerator Pedal with Driving from Superstructure (DSS)): Value 10 of parameter 13/12 changed to "not used"; Value 10 of parameter 13/14 changed to "not used" - Chapter 7.6.7.1. (Throttle inhibit via hardwired pin): Value 10 of parameter 13/12 changed to "not used"; Value 10 of parameter 13/14 changed to "not used" - Chapter 7.7.1. (DAI System Overview (TIER4i / EURO IV / EURO V)): Chapter name corrected - Chapter 7.7.2. (HW Input-/Output-Parameter): Binary Values 11/1 and 11/3 deleted; Input DI_FLEX_05 deleted; Value 2 of parameter 35/02 changed to "not used"; Parameters 13/27, 35/02, 35/21, 35/39 - Chapter 7.7.3. (TIER4i Inducement): Function of Pin3/10 changed to "Misc. (e.g. TIER4i DEF Lamp)" - Chapter 7.7.3.2. (Inducement Override) deleted - Chapter 7.7.4. (EURO IV / EURO V / Euromot 3b Inducement): Function of Pin3/10 changed to "Misc. (e.g. TIER4i DEF Lamp)" - Chapter 7.8.1. (Application of the Engine Brake): Example deleted - Chapter 7.8.2. (Activation of the engine brake systems): Abbreviation of transmission control unit changed to TCU; Disable conditions of engine brakes corrected - Chapter 7.8.4.1. (Anti-Lock Brake (ABS) Intervention and the Engine Brake): Function of pin 3/18 changed to "Misc (EnableAbsInput)"; Function of pin 4/13 changed to "Misc (EnableTempoSet)"; Value 10 of parameter 13/12 changed to "not used"; Value 10 of parameter 13/14 changed to "not used"
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		- Chapter 7.9. (Transmission Retarder): Transmission Retarder activation activation methods corrected - Chapter 7.9.1. (Application of the Transmission Retarder): Parameters 10/18, 10/19, 10/24, 10/25, 10/29, 10/31 added to parameter list - Chapter 7.9.3. (Transmission Retarder via hardwired connection): Function of pin 3/18 changed to "Misc (EnableAbsInput)"; Function of pin 4/13 changed to "Misc (EnableTempoSet)"; Fig. 7.27 corrected; Value 10 of parameter 13/12 changed to "not used"; Value 10 of parameter 13/14 changed to "not used" - Chapter 7.10.1. (Vehicle Speed): Value 10 of parameter 08/13 changed to "Output shaft speed from transmission (E-Can)" - Chapter 7.10.1.1. (Tachograph): Value 10 of parameter 08/13 changed to "Output shaft speed from transmission (E-Can)" - Chapter 7.10.1.2. (J1939 CAN message TCO1): Value 10 of parameter 08/13 changed to "Output shaft speed from transmission (E-Can)" - Chapter 7.10.1.3. (J1939 CAN message CCVS1): Value 10 of parameter 08/13 changed to "Output shaft speed from transmission (E-Can)" - Chapter 7.10.2. (Engine Speed): CAN connection between CPC4-ECAN and MR2 corrected to "E-CAN" - Chapter 7.10.8. (Air Filter Sensor): Function of output pin 3/10 corrected - Chapter 7.10.11. (Oil Pressure Lamp): Outputs for Service Brake Request Lamp deleted - Chapter 7.11.3. (Power Rating Curves): Value 2 of parameter 13/15 changed to "not used" - Chapter 7.12.1.3. (Idle Shutdown Override): One condition for idle shutdown override deleted - Chapter 7.12.2. (Engine Protection Shutdown): Function of pin 1/15 changed to "Shutdown override" - Chapter 7.13. (Fan): Function of pin 3/18 changed to "Misc (EnableAbsInput)"; Function of pin 4/13 changed to "Misc (EnableTempoSet)"; Value 10 of parameter 13/12 changed to "not used"; Value 10 of parameter 13/14 changed to "not used" SWR transmission retarder request eliminated - Chapter 8. (Diagnostics): Fig. 8.1 (Schematic illustration of the CPC4-ECAN Xentry DS communication) corrected - Chapter 8.1. (J1939 Diagnostic Messages): Abbreviation of transmission control unit changed to TCU - Chapter 8.2.2. (Fault Codes): Fault Codes SPN 625 FMI 16, SPN 630 FMI 2, SPN 630 FMI 9, SPN 630 FMI 14, SPN 3695 FMI 9, SPN 3695 FMI 13, SPN 3695 FMI 14, SPN 3695 FMI 19, SPN 3696 FMI 4, SPN 3696 FMI 9, SPN 3696 FMI 13, SPN 3696 FMI 14, SPN 3696 FMI 19, SPN 524285 FMI 4 deleted; Fault Codes SPN 158 FMI 2, SPN 247 FMI 0, SPN 247 FMI 1, SPN 247 FMI 9, SPN 247 FMI 10, SPN 247 FMI 14, SPN 625 FMI 2, SPN 625 FMI 8, SPN 625 FMI 9, SPN 625 FMI 10, SPN 625 FMI 13, SPN 625 FMI 14, SPN 625 FMI 15, SPN 625 FMI 19, SPN 630 FMI 13 corrected to MR2 control unit - Chapter 8.3.1. (Analog Values): Value 68 of analog value 48 changed to "not used"; For name and description of analog value 49 "MCM" replaced by "MR2"; Value 3 of analog value 60 changed to "not used"; Value 12 of analog value 66 changed to "TCU Min Speed"; For values 26, 27, 28 of analog value 68 "TCM" replaced by "TCU"; For values 9, 10 of analog value 69 "TCM" replaced by "TCU"; For values 8, 9 of analog value 70 "TCM" replaced by "TCU"; Values 10, 11 of analog value 72 changed to "not used"; For values 3, 7, 9 of analog value 77 "MCM" replaced by "MR2"; For name of analog value 77 "MCM" replaced by "MR2"; For values 13, 14 of analog value 94 "TCM" replaced by "TCU"; For value 14 of analog value 110 "TCM" replaced by "TCU"; Value 43 of analog value 120 changed to "diagnostic error MR2"; For value 5 of analog value 131 "TCM" replaced by "TCU" - Chapter 8.3.2. (Binary Values): Binary values 8/3, 8/4, 11/1 deleted - Chapter 8.4.1. (Routines for CPC4-ECAN): Value 12 for routine 17 changed to "not used"; Value 36 for routine 17 changed to "Tier4i DEF Lamp"; Value 37 for routine 17 changed to "Urea Low Lamp"; Values 38, 41, 50 for routine 17 changed to "not used"; Value 43 for routine 17 changed to "Eu4/5 Limitation Active Lamp"; Value 44 for routine 17 changed to "Tier4i Limitation Active Lamp"; Value 12 for routine 18 changed to "not used"; Value 36 for routine 18 changed to "Tier4i DEF Lamp"; Value 37 for routine 18 changed to "Urea Low Lamp"; Values 38, 41, 50 for routine 18 changed to "not used"; Value 43 for routine 18 changed to "Eu4/5 Limitation Active Lamp";
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		Value 44 for routine 18 changed to "Tier4i Limitation Active Lamp"; Value 12 for routine 19 changed to "not used"; Value 36 for routine 19 changed to "Tier4i DEF Lamp"; Value 37 for routine 19 changed to "Urea Low Lamp"; Values 38, 41, 50 for routine 19 changed to "not used"; Value 43 for routine 19 changed to "Eu4/5 Limitation Active Lamp"; Value 44 for routine 19 changed to "Tier4i Limitation Active Lamp"; • Chapter 9.3. renamed to "J1939 CAN Messages" • Chapter 9.3.3. (Default source addresses): Source addresses CM1DPF and DPFCM1 deleted; Abbreviation of Transmission changed to "TCU" • Chapter 9.3.4.1. (Miscellaneous messages): Source addresses CM1DPF and DPFCM1 deleted; Abbreviation "TCU" used for Transmission • Chapters 9.3.5. (CAN receive message details) and subchapters combined with chapter 9.3.6. (CAN send message details) and subchapters and shifted one level up to chapter 9.4. (CAN Message Details) • Chapter 9.4.1. (TSC1 – Torque/Speed Control 1 (to Engine)): Abbreviation "TCU" used for Transmission • Chapter 9.4.2. (TSC1 – Torque/Speed Control 1 (to Retarder)): Abbreviation "TCU" used for Transmission • Chapter 9.4.6. (CM1 – Cab Message 1 (receive)): Sender CM1DPF and DPFCM1 deleted • Chapter 9.4.13. (ESS – Proprietary A (Engine Start/Stop)): Abbreviation "TCU" used for Transmission • Chapter 9.4.18. (ETC1 – Electronic Transmission Controller 1): Abbreviation "TCU" used for Transmission • Chapter 9.4.22. (ETC2 – Electronic Transmission Controller 2): Abbreviation "TCU" used for Transmission • Chapter 9.4.25. (ETC7 – Electronic Transmission Controller 7): Abbreviation "TCU" used for Transmission • Chapter 9.4.26. (TCFG2 – Transmission Configuration 2): Abbreviation "TCU" used for Transmission • Chapter 9.4.30. (SOFT – Software Identification): Data Length corrected to 64 • Chapter 9.4.63. (PropB_51 – Proprietary B_51): DPF Lamp Status, DPF Status, DPF Regeneration Time marked als not supported; Correction of Byte and Bit orderings and signal names • Chapters 9.3.7. (Diagnostic CAN receive message details) and subchapters combined with chapter 9.3.8. (Diagnostic CAN send message details) and subchapters and shifted up one level to chapter 9.5. (Diagnostic CAN message details) • Chapter 9.5.2. (DM1 – Diagnostic Message 1, Active Diagnostic Trouble Codes (receive)): Abbreviation "TCU" used for Transmission
2018-04-12	1.1	• Chapters Application and Diagnostics added
2018-03-23	1.0	• Initial incomplete version (chapters for Application and Diagnostics missing)

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